# LensJudge: An Agentic Visual-Inspection Grader for Strong-Lens Candidates on the Claude Agent SDK

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June 2026

*From the Visual-Inspection Wall (v1) to Contaminant-Aware Escalation (v2)*

**Abstract**

The Huang group's ResNet/EfficientNet finders emit thousands of above-threshold cutouts per data release; turning them into graded candidates (A/B/C/D) is a human visual-inspection (VI) bottleneck. We build **LensJudge** — the system the group's onboarding report calls "LensAgent"/"VI Pre-grading Agent", renamed to avoid a name clash with an existing arXiv paper — on the *Claude Agent SDK for Python*, and report two generations. **Part I (v1)** asks whether the data on hand can *test* such a grader, and returns a sharp, load-bearing negative. The agent is strong where the task has clean physical features and confirmed labels — the *spectroscopic* discordant-redshift channel (20/20 Hsu Table-2 Grade-A pairs plausible, 4/4 Foundry-II non-lenses rejected) — but on *imaging* grading of the CNN's hard high-score pool *no* configuration (lean Sonnet, lean Opus, a judge panel, the §3 multi-agent design, a GIGA-Lens model fit, or hand-engineered representations) beats chance against the single-consensus A/B/C labels (AUC 0.32–0.51); a crossmatch shows why, with 53% of DESI grade-C candidates re-graded A/B by Euclid at 0.1″. The wall is *resolution* and *soft single-consensus labels*, not the grader — so the recommendation is to deploy as a high-recall *detect/escalate* pre-filter and reserve fine grading for high-resolution or confirmed data. **Part II (v2)** builds exactly that. A failure-mode rubric targeting the dominant LRG+companion/ring/spiral/blend mimics lifts lens-vs-*mimic* ranking from chance to 0.714 AUC (+0.196 over v1) while detection barely moves; a two-tier DESI→Euclid 0.1″ escalation, live-validated, flips 140 ambiguous cross-matches from DESI-undecided to Euclid-confirmed (median $p_{\text{lens}}$ 0.03 → 0.70, ∼90% agreement with the Euclid experts on grade-A); and an agency ablation shows the agentic loop is *unnecessary* for detection (a loop-free grader matches it ∼6× cheaper) but *essential* for mimic adjudication — motivating a cascade deploy (∼\$170 per 5k survivors). We are explicit about what stays unsolved: the gains are in ranking, not tight-FPR recovery; the fine A/B/C grade remains irreducibly soft even at 0.1″; and the escalated objects are independently re-graded catalogue lenses (a cross-validation), not new discoveries.

## 1 Motivation

Lens finding in wide-area imaging is a two-stage pipeline: a CNN ranks $O(10^7)$ galaxy cutouts, and humans visually grade the $O(10^3–10^4)$ above-threshold survivors into A (clear lens), B (probable), C (possible), and D (rejected) (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024). The VI step is the wall-clock bottleneck and will only worsen at Rubin scale. LensJudge automates the structured judgment of that step with a tool-using LLM agent, pre-grading candidates and escalating the genuinely ambiguous ones to a human — it is a triage assistant, not an autonomous arbiter.

**Two generations.** This report has two parts. **Part I** (§§3–11) builds the grader and *tests* it, and returns a load-bearing negative: imaging grading of the CNN's hard pool is near-chance and stays there under every backbone, deliberation design, and physical check we try — the binding constraint is ground-based *resolution* and *soft single-consensus labels*, not the agent. The actionable reading is to deploy as a high-recall *detect/escalate* pre-filter and reserve fine grading for high-resolution or confirmed data. **Part II** (§13) builds precisely that prescription — a contaminant-aware rubric, a two-tier high-resolution escalation, and an agency ablation that fixes the deploy cascade — and measures what each buys. The two parts share the data (§2), the architecture (§3), and the LensBench-VI harness (§4).

## 2 Do we have enough data to test it?

**Imaging: yes, today.** From the group's reproductions we assemble (all public / on-disk): *positives* — 2,706 graded A/B/C candidates as $101 \times 101 \times 3$ *grz* FITS (Storfer DR9 + Inchausti DR10) with per-candidate CNN probabilities; *hard negatives* — 2,345 Grade-D human-rejected candidates (the false positives the CNN flagged but humans rejected) parsed from the published catalogs, cutouts fetched on demand; *easy negatives* — ∼65K random-galaxy cutouts; *gold* — 21 Foundry-II confirmed/known lenses + 4 confirmed non-lenses (Huang et al., 2025). **Spectroscopy: catalog-level today.** Hsu's 13,530 discordant-redshift fiber pairs and 20 Table-2 Grade-A systems (Hsu et al., 2025) give a spectroscopic test set, though the raw DESI fiber flux is not local.

**Two honest gaps — one now partly closed externally.** (1) Every *DESI* grade is a *single* 2-senior-author consensus — there are no per-rater labels in the DESI catalogs — so a human-vs-human Cohen-$\kappa$ ceiling cannot be computed from them. We later locate external data that re-anchors a subset (§10): 291 candidates carry higher-resolution imaging, multi-grader grades, or spectroscopic confirmation, and we stand up the Euclid Q1 ($0.1''$, ∼10-vote) field as a high-resolution benchmark. A per-rater $\kappa$ over the *DESI* grades still needs the ∼250-cutout in-house study. (2) The spectroscopic grader reasons over catalog features ($z_l, z_s$, separation, $\sigma_v$, $\theta_\mathrm{E}$) plus DR10 imaging; raw-flux reading (SPARCL / public healpix coadds) is a documented enhancement.

## 3 Architecture on the Claude Agent SDK

The SDK has no image-in-prompt, so the only way the model sees pixels is a tool that returns an image content block. Our `fetch_cutout` tool renders the *grz* cube to multi-view Lupton-RGB PNGs (full / center-zoom / lens-light-residual / high-contrast, identical params to the group's inspection viewer) and returns them; `get_photometry` measures aperture $g-r, r-z$ colors; `crossmatch_local` flags prior-catalog overlap; `get_specfit` does the SIS $\theta_\mathrm{E}$/separation check. Tools are an in-process `create_sdk_mcp_server` bundle; because the Python `@tool structuredContent` is not forwarded, the grade contract is a system-prompt JSON schema validated into Pydantic models with one repair retry; PreToolUse/PostToolUse hooks stream a per-candidate JSONL trace (with `total_cost_usd`) that doubles as the eval harness.

We implement three imaging graders of increasing cost: **lean** (one multimodal call; the five Huang-2020 criteria as scored fields of a single integrated judgment), **panel** (four perspective-diverse judges — Advocate, Skeptic, Morphologist, Contaminant-hunter — fused by ordinal quorum + skeptic-veto + a 2-of-N A-rule), and **multiagent** (the onboarding-plan §9.1 factor-decomposition: morphology/color/ contaminant specialists → GradeArbitrator). A **spectro** grader classifies discordant-redshift pairs lens/dimple/not_lens.

# Part I

# Part I — v1: the visual-inspection wall

## 4 LensBench-VI methodology

The held-out set is a frozen, hash-pinned manifest of 200 systems: graded A/B/C positives + Grade-D hard negatives + random-galaxy negatives + Foundry-II gold, stratified by grade and region, negatives kept as separate strata. The LLM grader never trained on any cutout, so all rows are valid held-out for it (the NeuraLens training-set leakage that affects the frozen-CNN baseline is recorded per row). Tasks/metrics: (i) binary lens(A/B/C) vs non-lens(D) — ROC-AUC and recovery at 1%/0.1% FPR (the group's operating point); (ii) ordinal A/B/C/D vs consensus — quadratic-weighted $\kappa$, exact/adjacent accuracy, confusion; (iii) calibration of $p_{\text{lens}}$ (ECE, Brier); (iv) agent-vs-CNN agreement. Baselines: the $p_{\text{meta}}$ threshold, single nets, majority-class, and image-only-no-tools.

## 5 Results

On the full 200-system clean-imaging held-out set the lean Sonnet grader scores binary ROC-AUC 0.51, recovery@1% FPR 0.04, quadratic-weighted $\kappa = -0.02$, ECE 0.42 at \$0.06/candidate — i.e. essentially no agreement with the single-consensus A/B/C labels. Table 1 asks whether a stronger backbone or multi-agent deliberation recovers it, on a shared 48-system slice. All numbers are **consensus-referenced, no human ceiling**.

**Table 1:** Grader comparison on the shared 48-system imaging slice (consensus-referenced; no human ceiling). No configuration separates from chance.

| Metric | lean-Sonnet | lean-Opus | panel | multiagent |
|---|---|---|---|---|
| Binary ROC-AUC | 0.44 | 0.32 | 0.42 | 0.38 |
| Quad-weighted $\kappa$ | -0.13 | -0.06 | -0.07 | -0.13 |
| Mean cost / candidate (\$) | 0.06 | 0.13 | 0.25 | 0.20 |

**Key finding — on the hard imaging pool, neither the agent nor the CNN separates, and more compute does not help.** The frozen ensemble cannot separate real lenses from the Grade-D human-rejects (mean $p_{\text{meta}} \approx 0.82$ on rejects, near its value on lenses) — but, crucially, *nor does the agent*: every configuration sits at AUC 0.32–0.51 and $\kappa \approx 0$, assigning $p_{\text{lens}} \approx 0.02$–$0.03$ even to consensus-A systems, and a 2–4× more expensive Opus / panel / multi-agent run does not improve it. The agents are *uniformly skeptical* of the whole high-$p_{\text{meta}}$ pool.

**Why this is the right answer, not a bug.** We verified the cutouts are real and correctly resolved (all on disk; obvious arcs, e.g. `DESI-091.7214−58.9787`, do get high $p_{\text{lens}}$). The pool is the hardest possible slice — positives and CNN-rejected negatives *both* have high $p_{\text{meta}}$— and on it the A/B/C labels are mostly *unconfirmed* candidates whose C-vs-D boundary is one rater's subjective call. So an agent that is skeptical of unconfirmed C-grade candidates is not demonstrably wrong; it simply cannot be *adjudicated* against a single-consensus label. This is precisely what the missing multi-grader ceiling (and the absence of spectroscopic confirmation for most candidates) prevents us from resolving — the gap is the result. Appendix A shows three of these grading episodes end-to-end — a consensus-A lens, an ambiguous grade-C, and a Grade-D human-reject — abbreviated but otherwise verbatim, to give a concrete feel for how the agent reasons and uses its tools.

## 6 Reasoning made visible: Sonnet 4.6 vs Opus 4.8

Two instrumentation upgrades motivate this section. First, every grader call now *records the model's reasoning*: the Agent SDK's adaptive-thinking summaries are written into the per-candidate trace logs alongside the existing tool events, and the worked examples in Appendix A display them inline. Second, with reasoning observable we re-ran the lean grader on two backbones — `claude-sonnet-4-6` and `claude-opus-4-8` (1M context) — holding the rubric, tools, 6-turn limit and per-candidate budget cap fixed (the cap was raised to \$1.00 for Opus, whose episodes are costlier). Two caveats frame everything below: the displayed text is the API's *summarized* reasoning, not the raw chain of thought; and instrumenting capture revealed that adaptive thinking is the harness *default* — the frozen Table 1 runs were already reasoning silently. The comparison is therefore "reasoning made visible + backbone swapped", not "reasoning switched on".

Table 2 scores all three configurations on the identical 225-row manifest (the frozen baseline is re-scored on the full row set, which is why its numbers differ slightly from the $n$=200 clean-slice values quoted in §5).

**Table 2: Backbone comparison with reasoning captured** on the full 225-row LensBench-VI manifest (consensus-referenced; no human ceiling). "Baseline" is the frozen Table 1 lean-Sonnet run re-scored on the identical rows; its reasoning was on by default but unrecorded.

| Metric | lean-Sonnet (frozen baseline) | Sonnet 4.6 reasoning captured | Opus 4.8 (1M) reasoning captured |
|---|---|---|---|
| Binary ROC-AUC | 0.48 | 0.46 | 0.44 |
| Quad-weighted $\kappa$ | -0.08 | -0.10 | -0.03 |
| Recovery @1% FPR | 0.04 | 0.01 | 0.01 |
| Calibration ECE | 0.46 | 0.47 | 0.58 |
| Escalation rate | 0.48 | 0.42 | 0.12 |
| Mean $p_{\text{lens}}$ on truth-A | 0.13 | 0.14 | 0.05 |
| JSON parse rate | 100% | 100% | 100% |
| Mean cost / cand. (\$) | 0.06 | 0.08 | 0.15 |
| Mean wall (s) | 43 | 47 | 33 |
| Mean thinking chars | — | 3,359 | 1,647 |

**Imaging: the wall is real, reproducible, and backbone-independent.** The Sonnet re-run lands within noise of the frozen baseline (AUC 0.46 vs 0.48, $\kappa$ -0.10 vs -0.08) — the near-chance imaging result of §5 is stable run-to-run, not an artifact of one sample. Opus 4.8, at roughly 2× the per-candidate cost, does not buy agreement with the consensus labels either; it is the most decisively skeptical grader yet. It did not assign a single A grade on the 225-row manifest, graded 26 of the 30 consensus-A systems D, held $p_{\text{lens}}$ flat at ∼0.05 across all four truth grades, and escalated only 0.12 of cases (versus 0.42 for Sonnet) — where Sonnet hedges and hands borderline cases to a human, Opus rejects them with confidence. On the appendix's consensus-A system (`DESI-091.7214−58.9787`) the two backbones split exactly where the label is softest: Sonnet graded B/$p_{\text{lens}}$=0.76 three times running, Opus C/0.12. Curiously, Opus also produces *shorter* summarized reasoning than Sonnet (1,647 vs 3,359 characters per candidate) and finishes faster (33 s vs 47 s) — its cost premium is price per token, not verbosity.

**Spectroscopy: the gold-set result survives, with a twist.** With reasoning captured, Sonnet re-grades 19/20 Hsu Table-2 pairs plausible and rejects 4/4 Foundry-II non-lenses (the frozen run scored 20/20 and 4/4); it accepts 17/20 of the Foundry-II *confirmed* lenses. Opus roughly matches it on Hsu (18/20) and on the non-lenses (4/4), but accepts only 14/20 confirmed lenses — six reasoned rejections versus Sonnet's three. (A first Opus pass lost three episodes to JSON parse failures; we added the imaging path's one-shot repair retry to the spectro grader, after which all 44 parse and three of the apparent rejections resolved to acceptances.) The remaining rejections are not noise — Opus applies stricter physical-plausibility checks and catches real data-quality gaps, e.g.:

**Opus 4.8 — rationale for rejecting a confirmed lens (excerpt)**

```
theta_E is only 0.26" (sigma_v=152 km/s is modest), yet the imaging shows the companion ~3-4" north of the
     lens --- roughly 10x the Einstein radius, far outside any lensable geometry (the specfit 'strongly
     lensable' verdict is an artifact of the missing/zeroed separation). [...] Despite the discordant
     redshifts, this is a chance projection of two galaxies, not a strong lens [...]
```

On systems whose catalog features are incomplete, that stricter prior errs toward rejection — and on *confirmed* lenses, skepticism is not free. The stronger backbone buys better-argued verdicts, not better agreement.

For a qualitative feel for the two reasoning styles, compare the Sonnet episodes in Appendix A with this excerpt from the Opus episode on the same consensus-A system (it called `get_photometry`, which none of the three Sonnet recordings did):

**Opus 4.8 thinking (summarized) — DESI-091.7214−58.9787, excerpt**

```
The photometry suggests the annulus is bluer than the core, which weakly supports the first criterion, but
     the morphology is really what I need to focus on. [...] Critically, there's no blue tangential arc, no
      ring, and no symmetric counter-image pair. The greenish diffuse light around the core isn't actually
     blue and doesn't have an arc shape, so it doesn't fit the lensing morphology I'd expect.
[...] This sits right at the C/D boundary. The blue excess in the annulus and low-surface-brightness
     diffuse light near the lens point toward C (possible lens with a blue neighbor at plausible separation
      but weak morphology) [...] I'll set p_lens to 0.12, confidence to 0.6, leave contaminant null since
     this doesn't fit a clear artifact category, and mark it for human eyes given the ambiguity.
```

### 6.1 A prompt ablation: absolute evidence vs. relative triage

Opus's behavior suggests a hypothesis: the rubric states near-absolute evidence standards, and the more literal model applies them as written — while the human graders it is scored against were doing *relative triage* within a pool the CNN had already selected. We test this with a second rubric (`prompts/rubric_imaging_triage.md`) that holds the five criteria, tools, and output schema fixed and changes only the framing: grades are follow-up priority bins relative to the pool at $1.3''$ seeing; a textual "house scale" describes what consensus A/B/C/D actually look like at this resolution; a D *requires* a named contaminant; and the escalation policy is made asymmetric ("cannot confirm or refute" routes to C + escalate, not to D). $p_{\rm lens}$ is explicitly kept as an *absolute* probability. After a pilot on the shared 48-system slice confirmed the effect, both rubrics ran on the full 225-row manifest, whose strata let us score behavior on *hard* labels too: the 21 Foundry-II spectroscopically confirmed lenses ("gold") and the 30 random-galaxy negatives.

The split is clean: **the routing policy is prompt-fixable; the discrimination is not**. Under the triage rubric Opus's operationally dangerous failure mode disappears — escalation recovers from 12% to 68%, the reject-everything D-rate falls from 88% to 63%, and five times as many consensus-graded candidates stay in the lens bins. The gold tier makes the stakes concrete *on hard labels*: with the original rubric Opus kept only 14% of spectroscopically *confirmed* lenses in a lens bin — it would have silently discarded six of every seven proven lenses — versus 33% under triage framing (Sonnet: 33% → 57%). Yet no configuration's ranking moves outside noise: every AUC sits at chance (0.44–0.51, $n$=225), Opus still assigns essentially no A or B grades, and its absolute $p_{\rm lens}$ on consensus-A systems barely moves (0.05 → 0.09 — exactly as instructed, since $p_{\rm lens}$ was

**Table 3: Rubric ablation on the full 225-row manifest.** "Original" rows are the Table 2 runs. Recovery$_{ABC}$ = fraction of consensus-A/B/C systems kept in a lens bin; Gold = fraction of the 21 *confirmed* lenses kept in a lens bin (hard labels, not consensus).

| Config | Escal. | D-rate | Recovery$_{ABC}$ | Gold | mean $p_{\text{lens}}$ (truth-A) | AUC |
|---|---|---|---|---|---|---|
| lean-Sonnet (frozen) | 0.48 | 0.46 | 0.52 | 0.38 | 0.13 | 0.48 |
| Sonnet 4.6, original | 0.42 | 0.53 | 0.42 | 0.33 | 0.14 | 0.46 |
| Sonnet 4.6, triage | 0.70 | 0.35 | 0.66 | 0.57 | 0.15 | 0.51 |
| Opus 4.8, original | 0.12 | 0.88 | 0.08 | 0.14 | 0.05 | 0.44 |
| Opus 4.8, triage | 0.68 | 0.63 | 0.40 | 0.33 | 0.09 | 0.49 |

kept absolute). The framing also has a price that confirms it adds no information: Sonnet under triage rejects only 40% of the random-galaxy negatives that the frozen baseline rejected at 63% — the rubric shifts each model's operating point along the same recall/precision curve rather than lifting the curve. Prompting changes what the agent *does* under uncertainty (hedge and hand to a human, rather than reject with confidence); it cannot conjure separability that the $1.3''$ pixels do not contain. That is the same wall, now isolated to the data: the fix is better pixels (§10), not a better prompt and not a bigger model.

The three worked-example systems of Appendix A (recorded under the *original* rubric) trace the ablation in miniature. The consensus-A lens is framing-stable (Sonnet B/$p_{\text{lens}}$=0.75 original vs. B/0.74 triage; Opus C either way). The borderline grade-C system is exactly where the policy bites: Sonnet flips D $\rightarrow$ C with escalation under triage, and Opus, still grading D, now escalates instead of silently discarding. The Grade-D merger is rejected without escalation under all four configurations — a confident rejection of a *nameable* contaminant is robust to framing, which is precisely the behavior the asymmetric rule is meant to protect.

# 7 Spectroscopic pre-grading

On the gold spectroscopic set the catalog+imaging grader re-grades 20/20 of the 20 Hsu Table-2 Grade-A pairs as plausible lenses and rejects 4/4 of the 4 Foundry-II confirmed non-lenses, using the SIS $\theta_{\text{E}}$/separation consistency and the DR10 imaging.

# 8 The Foundry-I modelability criterion

The graders so far apply the Huang-2020 discovery rubric (imaging) and Hsu-2025 + physical-consistency reasoning (spectroscopy). The one Foundry-paper criterion runnable on a grz cutout is Foundry-I's (Huang 2025a): *does a plausible lens MODEL reproduce the configuration?* We implement it as `quick_lensmodel` — a real GIGA-Lens MAP fit (EPL+shear mass, two Sérsic lens-lights, Sérsic+shapelet source) run per cutout in the GIGA-Lens venv (~45 s, GPU, cached), keeping JAX out of the SDK process. The discriminator is a `lens_score` combining the recovered $\theta_{\text{E}}$ and the $\chi^2$ improvement from ray-tracing the source through the mass (large for a true lens, $\sim$0 for a smooth galaxy). The agent can call it as an extra tool (it does: verified end-to-end).

Crucially, evaluating it answers whether a *physical* check beats the vision graders on the hard pool — and it does not. On a balanced validation set the `lens_score` separates real lenses from *ordinary* galaxies well (AUC 0.77) but fails to separate them from the *hard Grade-D human-rejects* (AUC 0.44; the rejects' mean score is slightly *higher* than the lenses'). A held-out-slice run corroborates the pattern: `lens_score` is flat across graded-B/C, Grade-D rejects, and random galaxies. The interpretation is the same as for vision: Grade-D rejects are CNN-flagged objects that look lens-like enough to admit an EPL fit, so on ground-based *grz* a lens model reproduces them about as well as it reproduces true lenses. **Vision, a stronger backbone, multi-agent deliberation, and a physical lens-model fit all hit the same wall on the hard pool** — strong, convergent evidence that the limit is the data + the soft labels, not the grader. The modelability tool remains useful in its proper regime: filtering real lenses from ordinary galaxies (the broad finder pool), not from already-vetted hard rejects.

## 9 Engineered representations (RepresentationKit)

The graders so far feed the model the raw Lupton-RGB and let it infer the lensing geometry. We instead built a kit of physically-motivated transforms that make the signal explicit and emit scalar lensing-features (`common/representations.py`; the `lens_representations` tool): improved lens-light subtraction, a polar/tangential $(r, \theta)$ transform (an arc becomes a horizontal bar), a 180° symmetry residual (a counter-image pair lights up), a blue-excess color-isolation map at the Einstein radius, and a DoG+Hessian "arcness" ridge. Each feature is scored as a standalone discriminator (`eval/run_representations.py`) over a larger 545-system manifest with bootstrap CIs, on three contrasts.

**Table 4:** RepresentationKit standalone feature AUC (best single / logistic combo), with the agent rich-view arm. Consensus-referenced; no human ceiling.

| Contrast | best feature | combo |
|---|---|---|
| HARD: graded A/B/C vs Grade-D human-reject | 0.57 | 0.51 |
| EASY: graded A vs ordinary galaxy | 0.72 | 0.70 |
| GOLD: Foundry-II confirmed vs non-lens | 0.80(tangential/parity) | — |
| SILVER: noiseless sims, lensed vs unlensed | 0.80(arcness/asym.) | — |
| Agent (48-slice): lean RGB AUC 0.44 vs +representations 0.40 | | |

The kit closes the question cleanly. The features *demonstrably* extract lensing signal: on noiseless sims arcness/asymmetry reach AUC 0.80, and on spectroscopically *confirmed* labels the tangential-extent and counter-image-parity features reach 0.80. They also give a cheap easy-regime pre-screen (combo 0.70, ∼50 ms vs the 45 s GIGA-Lens fit). But on the *hard consensus pool* they collapse to combo 0.51 — and that does not change with cleaner Tier-2 implementations (photutils isophote, skimage Frangi, SEP geometry: all $\leq 0.57$) *nor* when the representations are fed to the agent (0.40 vs lean 0.44, at +43% cost). The *same features* score 0.80 on confirmed labels and 0.51 on the consensus pool — the sharpest evidence in this report that the hard-pool wall is a **label/data limit, not an algorithm limit**: vision, a stronger backbone, deliberation, a GIGA-Lens fit, hand-engineered representations (Tier-1 and Tier-2), and the agent reading those representations *all* land at chance on the soft Grade-D pool, while every method that is tested against clean (confirmed or simulated) labels succeeds.

## 10 External validation: do better data and more graders break the wall?

The RepresentationKit result localises the imaging wall to the labels and the $\sim 1.3''$ ground-based PSF. We test that directly: crossmatch the candidates against surveys that supply either *higher resolution* or *more graders*, then re-run the grader on the high-resolution pixels (`eval/crossmatch_external.py`, the external-archive workflow, and `eval/run_euclid.py`).

**The data exist.** Of the 4,354 unique candidates, **291 have external higher-quality data**: 104 in SuGOHI (HSC 0.6″, a ∼9-grader "committee" grade plus, for many, spectroscopic $z_l, z_s, \theta_{\mathrm{E}}$); 148 of the 1,414 grade-A and grade-B candidates have archival *HST* imaging (0.05–0.1″; ACS/WFC, WFC3) in MAST (99/481 grade-A, 49/933 grade-B); 77 are AGEL DR2 Keck-confirmed lenses; and 24 fall in the Euclid Q1 Strong Lensing Discovery Engine field (VIS 0.1″, ∼10 expert votes each). **89 of the 291 are DESI grade-C** — precisely the soft pool where the agent and the consensus are unsure.

**The "C" grade is information-limited at ground resolution.** For the 24 candidates that Euclid independently rediscovered, the DESI 1.3″ grade and the Euclid 0.1″ 10-expert grade disagree in one direction: **53% of the DESI grade-C candidates are graded A or B by the Euclid panel** (6 of 17 jump to grade A). The "C" was not a weak lens; it was an unresolved one.

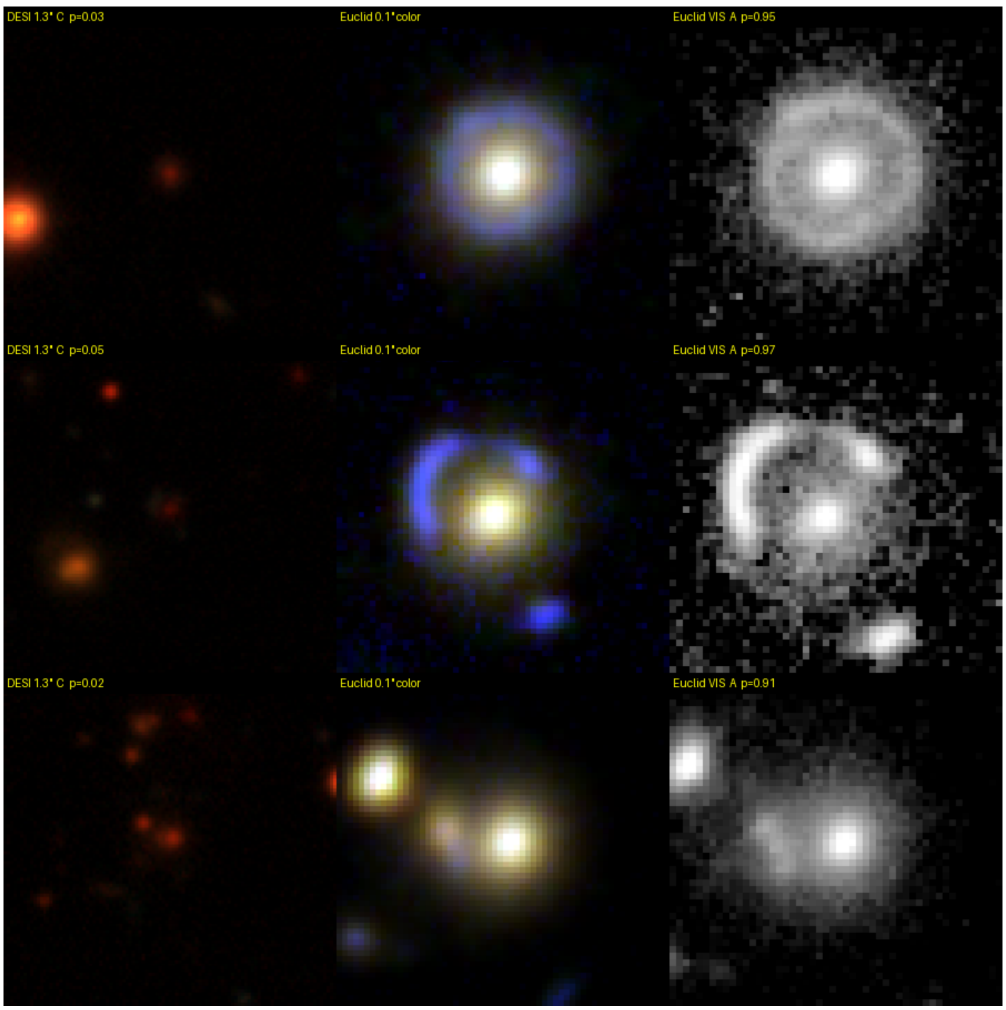



**Figure 1:** Three DESI grade-C candidates that fall in Euclid Q1. *Left*: DESI grz at 1.3″ seeing ($p_{\text{lens}}$ 0.02–0.05; the agent grades D). *Middle/right*: Euclid 0.1″ colour and lens-subtracted VIS — unmistakable Einstein rings/arcs ($p_{\text{lens}}$ 0.91–0.97; the agent and 10 Euclid experts grade A). Same agent, same rubric; only the pixels change.

**Resolution breaks the *detection* wall (paired, within-object).** For the 10 matched candidates with Euclid cutouts on disk we grade the *same object* twice — on the DESI grz cutout and on the Euclid VIS+NIR cutout (`fetch_euclid_cutout`, a $0.1''$ analog of `fetch_cutout`). Mean $p_{\rm lens}$ rises from 0.14 (DESI) to 0.75 (Euclid), increasing on 9 of 10; on the DESI grade-C subset it goes from 0.05 to 0.90. The objects the agent rejected as flat blobs at $1.3''$ it grades as clear rings at $0.1''$ (Fig. 1), agreeing with the Euclid experts. This is the within-object proof that the hard-pool wall was resolution- (and label-) limited, not algorithm-limited. Part II (§15) turns this paragraph into a deployable two-tier grader and scales it from these 10 hand-matched objects to 140 escalated cross-matches.

**But the fine A/B/C grade stays soft even at $0.1''$.** Ranking 95 Euclid candidates by $p_{\rm lens}$ against the 10-expert grade gives Spearman $\rho = 0.08$ (n.s.) and A-vs-C AUC = 0.51 — chance. This is the RepresentationKit finding again at 13× the resolution and 10× the graders: the binary *detection* question is resolution-limited and opens up with better data, but the *fine confidence grade* A/B/C is an irreducibly soft human judgment that resolution does not make reproducible. (The grade-C objects with Euclid cutouts are the modelable, lens-like C's, so this is a conservative contrast.)

## 11 The human ceiling is low — and now estimated

Every imaging number above is consensus-referenced because the DESI grades are a single 2-author consensus with no per-rater votes. Two independent routes now bound the human ceiling (`eval/human_ceiling.py`); per-classifier raw votes turn out to be unavailable publicly from any survey (Space Warps releases only skill-weighted SWAP posteriors; SuGOHI/Euclid release only aggregated consensus), so we use inter-*team* agreement plus the published literature.

**Inter-team agreement (computed, data in hand).** The crossmatch gives the same candidates graded independently by other expert groups, so their agreement estimates the ceiling. On a quadratic-weighted $\kappa$: DESI team vs the SuGOHI ∼9-grader committee = 0.29 (95% CI 0.12–0.44; $n = 103$); DESI team vs the Euclid ∼10-expert panel = 0.17 ($n = 24$; partly a genuine resolution effect, §10). Independent expert teams agree only "fairly" on A/B/C.

**Published human reliability (literature).** This matches the field. Rojas et al. (Rojas et al., 2023) (55 classifiers, 1489 images) find only 73% of repeat gradings identical (intra-rater, an *upper* bound; ∼QWK 0.8) and need $\geq 6$ graders to stabilise a marginal grade; Petrillo et al. (Petrillo et al., 2019) report that of candidates flagged by $\geq 1$ of 7 inspectors only 4.5% were flagged by all 7; Cañameras et al. (Cañameras et al., 2021) find only 30% of a blind set had zero dispersion among 5 graders; the Euclid Q1 engine (Euclid Collaboration et al., 2025) uses 10 experts/object yet its panel is decisive on only ∼38% of candidates. The A/B/C grade is intrinsically soft.

**Where the agent sits.** On the same SuGOHI-matched set at DESI resolution, LensJudge agrees with the DESI consensus at only $\kappa = 0.00$ ($n = 104$) — *below* the human inter-team ceiling (essentially uncorrelated with the consensus): it is uniformly over-skeptical on $1.3''$ cutouts, grading 66 of these 104 HSC-confirmed candidates "D" and none "A" because the arcs are unresolved. It reaches human-level only when given resolution that disambiguates the lens (the Euclid paired test, §10; $p_{\rm lens}$ 0.05 → 0.90). The operative limits, in order: (i) the fine A/B/C grade is barely reproducible by anyone ($\kappa \sim 0.3$ between expert teams, 73% intra-rater); (ii) at ground resolution the agent underperforms even that, over-rejecting blurred lenses; (iii) high-resolution data removes the ambiguity for the binary detection question. The actionable reading, now quantified: deploy LensJudge as a high-recall detect/escalate pre-filter and reserve the "grade" for confirmed or high-resolution data, rather than treating it as a target to reproduce.

## 12 Caveats and what is not done (Part I)

(1) **No per-rater human ceiling** — all DESI labels are single consensus. We now anchor a subset against external *multi-grader* grades (Euclid Q1 ∼10 votes; SuGOHI committee) and HST/Keck confirmation for 291 candidates, but a per-rater $\kappa$ still needs either the ∼250-cutout in-house study or Zooniverse per-classifier votes (consensus is all that is publicly downloadable). (2) **Grade-D negatives are lens-like** (CNN-flagged), so the binary AUC is lower — and more honest — than against random galaxies; strata are scored separately. (3) The **CNN pre-filter gate** (a cost lever at full-survey scale) is not exercised here because the eval set is by construction the high-$p_{\mathrm{meta}}$ ambiguous middle band. (4) Orchestration of the §9.1 specialists is *programmatic* rather than LLM-dispatched `AgentDefinition` subagents, for reliability. (5) Spectroscopic raw-flux streaming is a documented enhancement.

### Part II

# Part II — v2: contaminant-aware rubric, escalation, and the agency ablation

Part I localised the imaging wall to ground-based resolution and soft single-consensus labels and prescribed a deployment: a high-recall *detect/escalate* pre-filter, fine grading reserved for high-resolution or confirmed data. Part II builds that prescription as four concrete changes and measures what each one buys — a failure-mode rubric (§13), an honest detection-vs-grading benchmark (§14), a two-tier high-resolution escalation (§15), and an agency ablation that fixes the deploy cascade (§16). Throughout, v2 is a workstream of the sibling ClaudeNet contaminant-aware finder program (`reproductions/claudenet`); the lens-mimic negatives and the Euclid-overlap candidate lists below come from that program, and every number is read from a tracked artifact under `outputs/`.

## 13 The dominant failure mode and the contaminant-aware rubric

Part I's wall is about the *soft* grade-C/D middle, where no label is decisive. But a large and *nameable* slice of the imaging false positives is not soft at all: the **LRG+companion** (a luminous red galaxy with a nearby blob) is ∼half of the CNN's high-score contaminants in the ClaudeNet sweeps, and the v1 rubric — the five Huang-2020 *positive* criteria — never told the agent to rule it out. `prompts/rubric_imaging_v2.md` adds a "rule these out" section for the dominant mimics (LRG+companion, ring galaxy, spiral, blend, merger) keyed on two physically-grounded discriminators: *colour* (a real lensed source is *bluer* than the lens, a companion matches its colour) and *radial geometry* (a real arc is tangentially curved at fixed radius with a counter-image, a companion is round and one-sided). Aperture photometry is promoted from optional to often-decisive (the annulus-bluer-than-core test), and the contaminant enum gains `lrg_companion/blend`. This targets *lens-vs-mimic separation* — a different axis from Part I's lens-vs-consensus grading — so it needs its own benchmark, with the v1 rubric kept as the controlled baseline.

## 14 An honest detection-vs-grading benchmark

To score the rubric without conflating the two failure modes, the calibration harness (`eval/run_lensjudge_eval.py`) partitions a frozen 120-row evidence manifest (50 graded A/B/C + 40 lens-mimics + 30 random galaxies) into two benchmarks: **Benchmark A (detection)** — lens vs random galaxy — and **Benchmark B (separation)** — lens vs *mimic*, whose negatives are the ClaudeNet-v3 601-row mimic seed (CNN-high, agent-confirmed non-lenses), a *cleanly-labelled* contaminant class, unlike Part I's soft single-consensus Grade-D pool. A \$100 evidence gate requires a v2 change to beat the pinned v1-lean baseline on the benchmark before any bulk grading is paid for.

**Table 5:** Gated comparison, v1-lean vs v2 (rubric$_{v2}$ + escalate), on the frozen 120-row evidence manifest (sonnet; the pair cost \$16.6 total, parse 120/120). The v2 rubric lifts lens-vs-*mimic* ranking by +0.196 AUC while detection barely moves; it also rejects more mimics (D→D) and assigns lower $p_{\text{lens}}$ to non-lenses.

| config | Bench-A AUC (lens-vs-random) | Bench-B AUC (lens-vs-mimic) | mimics D→D (/40) | $p_{\text{lens}}$ (non-lens) mean | \$/cand |
|---|---|---|---|---|---|
| v1-lean | 0.593 | 0.518 | 47 | 0.12 | 0.068 |
| v2 (rubric$_{v2}$+esc.) | **0.668** | **0.714** | 53 | 0.07 | 0.071 |

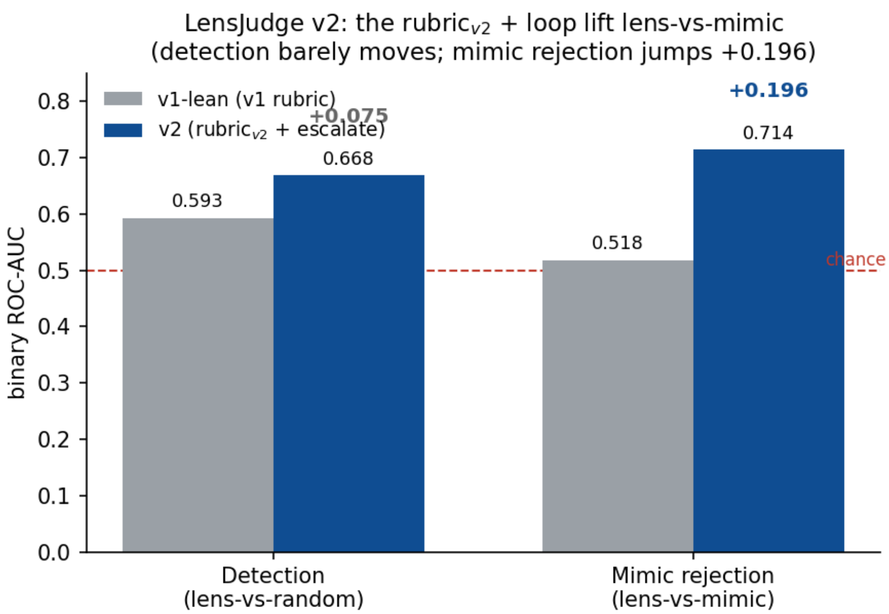



**Figure 2:** The v2 headline. Detection (lens-vs-random) barely moves (+0.075), but the failure-mode rubric lifts lens-vs-*mimic* ranking from chance to 0.714 AUC (+0.196). Source: `outputs/agency_ablation.json`.

**What this does and does not say.** It does *not* overturn Part I's wall. Part I's near-chance result is on the *soft* single-consensus Grade-D pool, where the label itself is a subjective call; Benchmark B's negatives are the *cleanly-labelled* v3 mimics, and the v2 gain is a *ranking* (AUC) effect — recovery at the group's tight 1%/0.1% FPR operating points stays ≈0 for *both* configs because the agent's $p_{\text{lens}}$ remains uniformly low (the documented over-skepticism at 1.3″; §11). What v2 fixes is *which mimics rank above which lenses*, the quantity the downstream finder cares about — not absolute calibration, and not Part I's irreducibly soft A/B/C grade.

# 15 Two-tier high-resolution escalation, live-validated

Part I (§10) showed the *detection* wall opens at 0.1″ but proved it on 10 hand-matched objects. v2 productizes it. `imaging/grader_escalate.py` grades tier-1 on DESI $grz$; when `common/highres.py:resolve_highres` finds Euclid (or, in future, HSC) coverage, it re-grades the *same object* at tier-2 0.1″. The trigger is anything *not* a confident tier-1 "A" — so the over-skeptical grader's buried lenses get the second look they need — and the real cost gate is *coverage* (a cheap local catalogue lookup), so on a full sweep only the rare Euclid-overlapping survivors ever pay for a tier-2 grade.

**Live validation.** On 6 Euclid-covered south objects ($0.71 total): the 3 grade-A lenses the DESI grader *missed* (tier-1 $p_{\rm lens}$ 0.02/0.03/0.40, mean 0.15) all flip to "A" (mean 0.96); critically it *discriminates* — the one true non-lens C correctly *stays* D (0.02), not inflated. **At scale**, across the v3 Euclid-overlap runs (D1 + D4 + D5 + the south validation = 140 escalated cross-matches), the median $p_{\rm lens}$ flips 0.03→0.70 from tier-1 to tier-2 (Fig. 3), and our tier-2 grades agree with the Euclid experts on ∼90% of the grade-A.

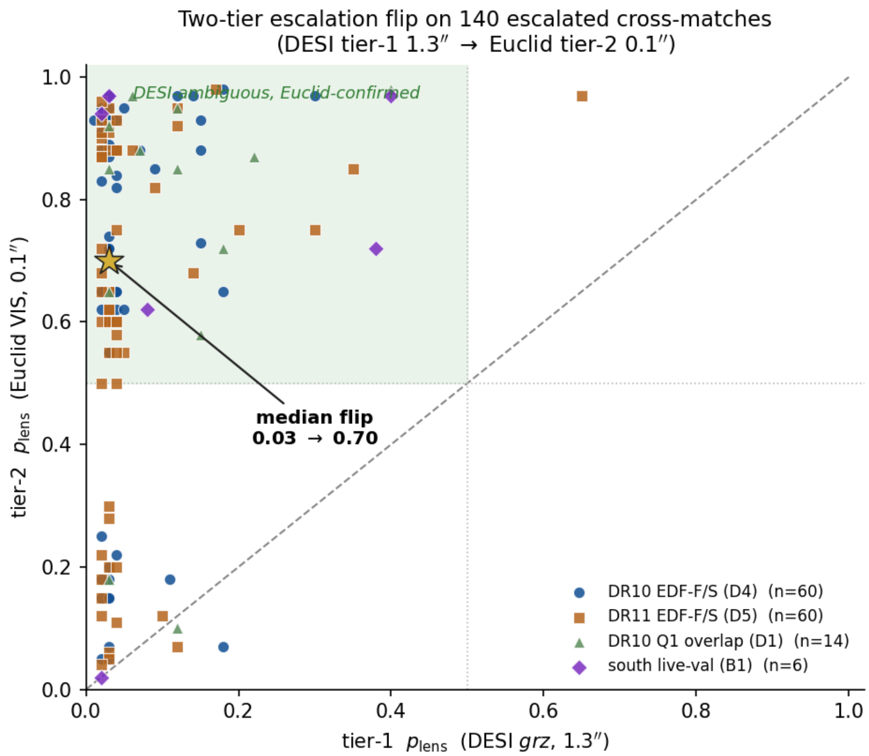



**Figure 3:** The escalation flip on 140 escalated cross-matches (the productized, quantitative successor to the 3-object montage of Fig. 1). Nearly every object is DESI-ambiguous (tier-1 $p_{\rm lens} \lesssim 0.2$) yet Euclid-confirmed (tier-2 $\gtrsim 0.5$); the star marks the median flip 0.03→0.70. Source: the four `*_vet.parquet` escalation runs.

**What it is, precisely.** These are independently re-graded objects from the published Euclid Q1 catalogue — a *cross-validation* of the DESI finder against Euclid, not net-new discoveries (the sibling ClaudeNet report makes the same statement about the same objects). And escalation breaks *detection*, not grading: the fine A/B/C grade stays soft even at 0.1″ (Part I's A-vs-C AUC 0.51), so tier-2 converts "unresolved blob" into "clear lens", not "C" into a reproducible "A".

## 16 Does the agentic loop contribute?

LensJudge's "agentic" surface is nearly a fixed pipeline: a `ToolSearch`, one `fetch_cutout`, a `get_photometry` call ∼38% of the time, then a grade. Does the tool-call *loop* earn its cost over invoking the same tools programmatically? `imaging/grader_direct.py` removes the loop entirely — it renders the same views + photometry in Python and makes *one* base Messages-API call with the images inline (no tools, 1 turn). The $2 \times 2$ (rubric $\times$ loop) plus a thinking arm, on the same 120 rows:

**Table 6:** Agency ablation (same 120-row evidence set, sonnet). The agentic loop is *free to skip* for detection but is the *only* arm that reaches the lens-vs-mimic gain — which therefore lives in the multi-turn loop, not the rubric text or a thinking scratchpad. Values from `outputs/agency_ablation.json`.

| arm | rubric | loop | Bench-A AUC | Bench-B AUC | $/cand | turns |
|---|---|---|---|---|---|---|
| lean | v1 | yes | 0.593 | 0.518 | 0.068 | 3.75 |
| direct | v1 | no | **0.669** | 0.384 | 0.012 | 1 |
| **v2** | v2 | yes | 0.668 | **0.714** | 0.071 | 4.42 |
| direct | v2-tool | no | 0.585 | 0.401 | 0.013 | 1 |
| direct | v2-inline | no | 0.659 | 0.407 | 0.012 | 1 |
| direct+think | v2-inline | no | 0.531 | 0.548 | 0.030 | 1 |

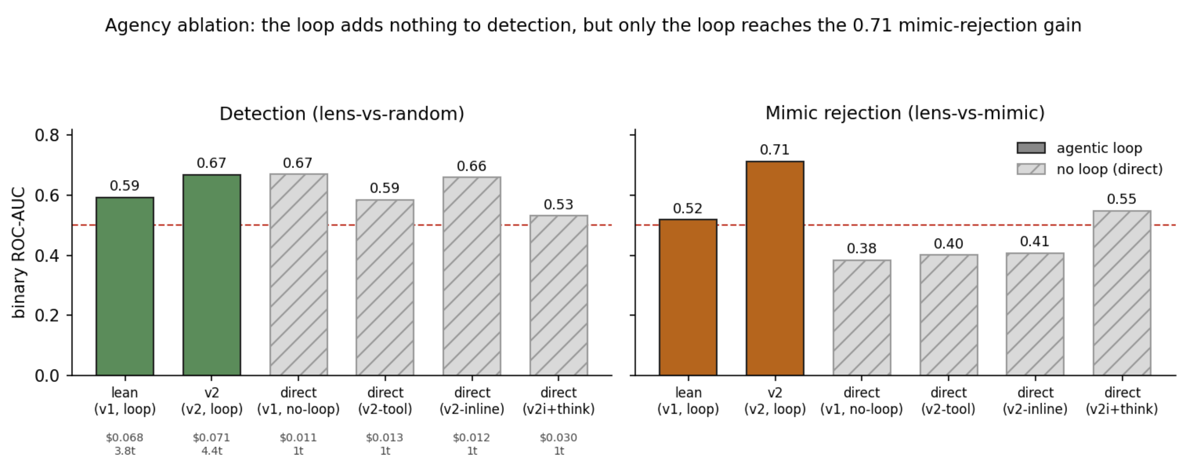



**Figure 4:** The two findings, by panel. *Left (detection):* every arm clusters near 0.5–0.67 — the loop adds nothing, and the loop-free `direct` grader matches it at ∼1/6 the cost. *Right (mimic rejection):* only the two loop arms clear chance, and only the v2-rubric loop reaches 0.714. Hatched bars are loop-free.

**(1) Detection: the loop adds nothing.** Loop-free `direct` matches/beats `lean` on lens-vs-random AUC (0.669 vs 0.593) at ∼1/6 the cost ($0.012 vs $0.068, 1 turn vs 3.75) — the tools are invokable programmatically with no detection loss. **(2) Lens-vs-mimic: the loop matters, and it is not the rubric text.** The mimic gain ($0.518 \to 0.714$) lives in the *loop* arm only: feeding the identical v2 content inline does not transfer it (0.407), and a thinking scratchpad recovers only half (0.548) while *hurting* detection. The multi-turn structure — inspect the cutout, then the photometry, reason across turns — is what does the hard mimic discrimination. **(3) Routing is degenerate.** The LLM grade-gate fires on ∼100% of candidates (over-skeptical), carrying no selectivity a free CNN $p_{\mathrm{meta}}$ threshold cannot reproduce (Jaccard 0.017); *coverage*, not the LLM, bounds tier-2 cost.

**Deploy as a cascade.** The ablation prescribes the stage-2 survivor-vetting pipeline directly: cheap `direct` triage for detection on the bulk, the agentic *v2* grader for mimic adjudication on the survivors that pass, and Euclid escalation gated by *coverage* (not the LLM grade). Per 5,000 survivors that is ∼\$60 (all-direct) / ∼\$170 **(cascade)** / ∼\$355 (all agentic-v2) / ∼\$1,575 (all-multiagent) — the cascade buys the mimic-adjudication gain for ∼2× the cheapest option and ∼9× less than naive multiagent.

**Caveats (v2).** The evidence set is $n$=120, so the AUCs are directional (CIs $\approx \pm$ 0.06–0.08) and the headline finding (2) is softer in magnitude than in direction; single seed, one model (sonnet), one manifest. `grader_direct` always computes photometry (a deliberate stance, not a bug; `lean` called it ∼38% of the time). The routing gate is degenerate, and the fine A/B/C grade is unmoved — Part I stands. Finally, B7 is a v2 *ablation*, not "LensJudge v3": that label denotes the sibling ClaudeNet program, and the cascade is the natural v3 milestone only *if* adopted as the deploy default.

## Part III

# Part III — v3: deeper resolution, the deployed cascade, and the active-learning loop

Part II ended with a prescription it had not yet built: deploy the grader as a *cascade* (cheap detection triage, agentic mimic adjudication, high-resolution escalation by coverage), and reach for more resolution than Euclid Q1's tiny DESI overlap supplies. Part III builds both, and feeds the result back to the finder. (Implemented on branch `lensjudge-v3`; every number below is read from a tracked artifact under `outputs/`.)

## 17 HSC PDR3 tier-2: widening the resolution lever

Euclid Q1 (0.1″) is the sharpest tier-2 source but overlaps only ∼24 DESI candidates (§15). **HSC-SSP Wide** (0.168″/px, ∼0.6″ seeing) covers far more equatorial sky — an intermediate rung between DESI's 1.3″ and Euclid's 0.1″. We add it as a second tier-2 source: `common/hsc_fetch.py` pulls grizy coadd cutouts from the public PDR3 `das_cutout` service (HTTP Basic auth, cached; a 404 = out of footprint = a clean no-op), and `resolve_highres` now tries `(euclid, hsc)` in priority, the escalate grader dispatching tier-2 by survey.

**Validation (the Euclid-B1 analog).** We grade the 26 *known* HSC lenses that cross-match our DESI candidates (SuGOHI/HOLISMOKES committee grades) through the escalate grader; 25/26 fall in the PDR3 footprint. The over-skeptical DESI grader buries them — tier-1 grades D:16 / C:8 / B:1, median $p_{\mathrm{lens}}$ 0.04 — and **HSC tier-2 recovers** 23/25 = 92% **as A/B**, median $p_{\mathrm{lens}}$ 0.04→0.62 (rose on 24/25), agreeing with the committee on 92% (Fig. 5). The 0.6″ flip is shallower than Euclid's 0.1″ (0.62 vs the ∼0.70 of §15), exactly as the coarser rung predicts; the 2/25 that stay D are subtle/blended even at HSC. Appendix B (B.1) shows one such recovery end-to-end. (Footprint-bounded — HSC Wide overlaps DESI only partially — and these are *known* lenses, a recovery test, not new discoveries.)

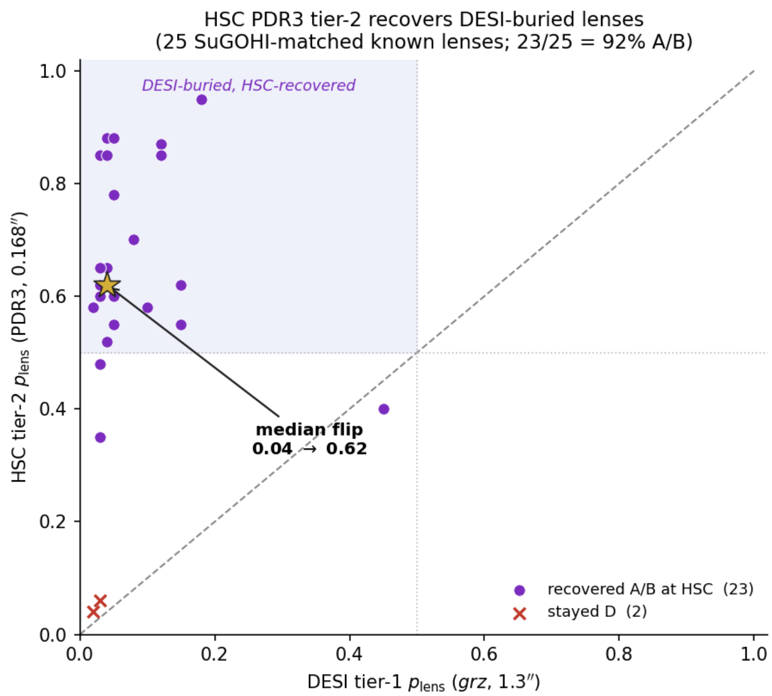



**Figure 5:** HSC PDR3 tier-2 on the 25 SuGOHI-matched known lenses with HSC coverage: the over-skeptical DESI grader buries them (tier-1 $p_{\rm lens}$ ∼0.04); HSC 0.168″ recovers 92% as A/B (median flip 0.04→0.62). The HSC analog of Fig. 3, one resolution rung coarser than Euclid.

## 18 The deployed cascade

`eval/run_cascade.py` productizes the B7 recommendation. Stage 1 runs the cheap loop-free `direct` grader on every survivor; stage 2 escalates the top `pass_frac` *by the stage-1 rank* to the agentic v2 grader (plus tier-2 by coverage). We route by *rank*, not the direct grader's absolute grade, because that grade is over-skeptical and over-confidently names contaminants — in Appendix B (B.1) stage 1 grades a *real* lens "D / lrg_companion" with 0.88 confidence — so its trustworthy signal is *ranking*.

**Does rank-routing keep the lenses?** On a 120-row labelled set (50 lens / 40 mimic / 30 random) the cheap stage-1 score has detection AUC 0.605, and rank-routing behaves as designed (Fig. 6): **lens recall exceeds the escalated fraction** at every operating point (66% at `pass_frac`=0.5) and **randoms drop well below it** (27%) — the cheap stage enriches lenses and discards obvious non-lenses. It does *not* separate lenses from *mimics* (mimic escalation ≈ base rate), which is exactly what the agentic stage 2 is for (the B7 division of labour). The cost is the measured B7 ladder: ∼\$60 (all-direct) / ∼\$170 (cascade) / ∼\$355 (all-v2) / ∼\$1,575 (all-multiagent) per 5k survivors. Honest cost: ∼34% of real lenses are dropped at stage-1 at `pass_frac`=0.5, so a high-recall deployment raises it.

## 19 Closing the loop: crossmatch and hard-negative export

Two smaller pieces complete the v3 increment. **Expanded crossmatch** (`eval/crossmatch_external.py`) adds SuGOHI (local) and best-effort HST/AGEL (VizieR) to the Euclid Q1 match, yielding 100 candidates with external high-resolution coverage (24 Euclid + 76 SuGOHI) — the anchor set for the HSC validation above. **Hard-negative export** (`eval/`

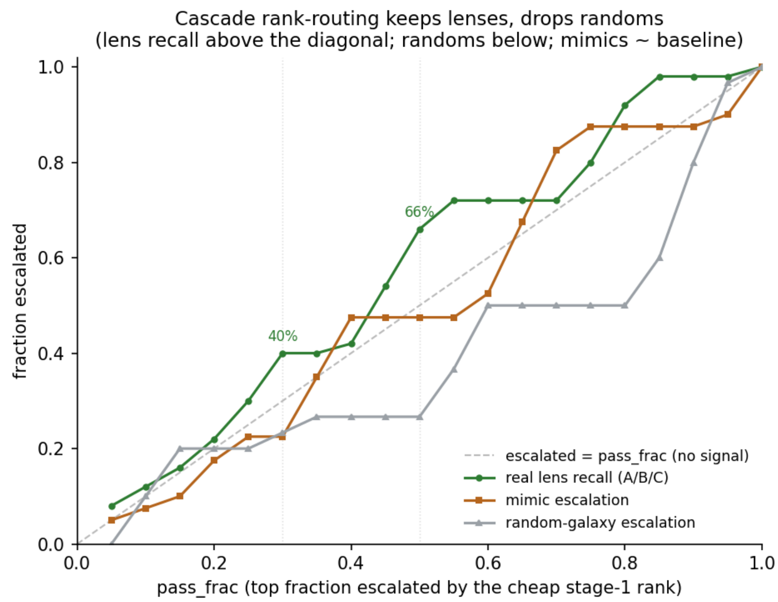



**Figure 6:** Cascade rank-routing on the 120-row labelled set: real-lens recall (green) sits above the no-signal diagonal, random-galaxy escalation (grey) below it, while mimic escalation tracks the baseline — the cheap stage-1 enriches lenses and drops randoms; lens-vs-mimic adjudication is left to the agentic stage-2.

`export_hard_negatives.py`) writes the agent-confirmed non-lenses (grade D + a named contaminant; e.g. Appendix B B.2) into the ClaudeNet-v3 mimic-bank schema — 129 rows tagged `source=lensjudge_v2` — closing the LensJudge → finder active-learning loop (folded in by the v3 bank assembler; we do not retrain here). A no-API test suite (`tests/test_v3.py`, 8/8) guards the graceful-fail, schema, and rank-routing logic.

# 20 Conclusion

LensJudge can be *built* on the Claude Agent SDK and, more importantly, *tested* on data in hand — and across two generations the test sharpened into a usable answer. **Part I** returned the load-bearing negative: imaging grading of the CNN's hard pool is near-chance against the single-consensus A/B/C labels, and *no* backbone, deliberation design, lens-model fit, or hand-engineered representation moves it — the binding constraint is ground-based *resolution* and *soft labels*, not the agent, with the actionable prescription to detect/escalate. **Part II** built that prescription and measured it. A failure-mode rubric plus the agentic loop recover lens-vs-*mimic ranking* from chance to 0.714 AUC (+0.196) on the cleanly-labelled mimic bank — the discrimination Part I found unbeatable on *soft* labels — and a two-tier escalation converts 140 resolution-limited cross-matches from DESI-undecided to Euclid-confirmed (median $p_{\mathrm{lens}}$ 0.03 → 0.70), agreeing with the experts. What is now *solved*: detection (via escalation) and mimic ranking (via the rubric + loop). What *remains*: the fine A/B/C grade is irreducibly soft even at 0.1″; the v2 gains are in ranking, not tight-FPR recovery; the escalation routing gate is degenerate; and the evidence $n$ is modest. **Part III** builds the deployment: a rank-routed cascade (cheap triage enriches lenses, the agentic loop adjudicates mimics) plus an HSC 0.6″ tier-2 that — validated on 25 known lenses — recovers 92% of the lenses DESI buries, widening the resolution lever beyond Euclid's sparse overlap, and it exports the agent's confirmed mimics back to the finder (closing the active-learning loop). The open frontier is the one Part I named: a per-rater human ceiling (the ∼250-cutout, ≥ 2-grader study) that would turn “ranking” into a claim about truth.

# A The agent in action: three worked examples

This appendix shows the lean single-agent grader (§3; the lean-Sonnet row of Table 1) grading three real LensBench-VI members end-to-end: a consensus Grade-A lens, an ambiguous Grade-C “possible”, and a Grade-D human-rejected contaminant. Episodes were recorded with full message-stream capture (`eval/run_appendix_examples.py`; the benchmark traces log tool events only) using the identical configuration scored in Table 1 — the *original* rubric of §3, not the triage variant of §6.1: `claude-sonnet-4-6` on the Claude Agent SDK, the `fetch_cutout` + `get_photometry` tool pair, a 6-turn limit and a $0.50/candidate budget cap (observed: $0.066–$0.147 and 31–59 s per episode). The agent’s *reasoning* is shown in “thinking” boxes at the position it occurred. Two caveats: the displayed text is the API’s *summarized* reasoning, not the raw chain of thought, and adaptive thinking is the harness default — the instrumented recordings in §6 make visible what earlier runs (including the Table 1 benchmark) performed silently. By design the lean grader is prompted to “respond with ONLY the JSON object”, so beyond the thinking blocks its deliberation surfaces in the tool-call sequence and in the structured `criteria/rationale` fields of its answer. Three editing rules apply: (i) abbreviation is by *elision only*, marked `[...]` — the agent’s words (thinking included) are never paraphrased; (ii) a harness-internal tool-discovery call at the start of each episode and the `mcp__lens__` tool-namespace prefix are elided, and repeated tool-output boilerplate is shown in the first episode only; (iii) the PNG views the tool returns to the agent are reproduced as figure strips (re-rendered pixel-identically from the same cutout cube). The three episodes were selected for legibility from 9 reasoning-instrumented recordings of the same three systems (none of which happened to invoke `get_photometry` — the Opus 4.8 episode excerpted in §6 shows that exchange); each example cross-references the grade the same system received in the frozen Table 1 benchmark run.

## A.1 Positive: a consensus Grade-A lens

`DESI-091.7214−58.9787` — consensus grade A, CNN $p_{\mathrm{meta}} = 0.9998$; the benchmark lean run graded it B with $p_{\mathrm{lens}} = 0.75$.

**task message**

```
Grade this strong-lens candidate. name='DESI-091.7214-58.9787', survey='storfer'. RA=91.721434, Dec
     =-58.978734
Tractor type: DEV, region: DECaLS.
CNN ensemble scores (prior, not ground truth): p_meta=1.000.

Call fetch_cutout with that name and survey, inspect the views, then respond with ONLY the JSON object.
```

**thinking (summarized)**

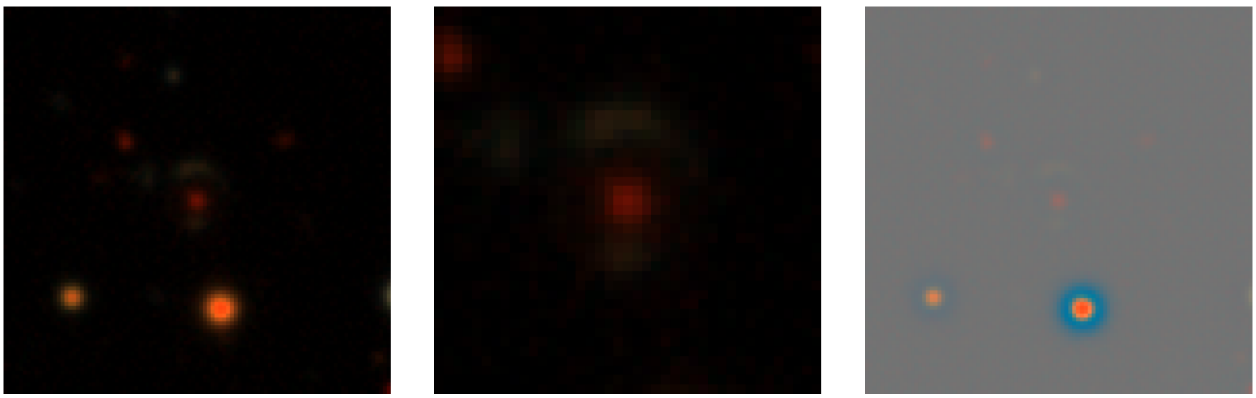
```
The user wants me to grade a strong-lens candidate. I need to:
1. Call ‘fetch_cutout‘ to get the views
2. Optionally call ‘get_photometry‘ and other tools
3. Return a JSON grade

Let me first load the schema for ‘fetch_cutout‘ and ‘get_photometry‘ tools.
```

**tool call — fetch_cutout**

{"name": "DESI-091.7214-58.9787", "survey": "storfer", "ra": 91.721434, "dec": -58.978734, "views": ["full", "zoom", "residual"]}

**tool result — fetch_cutout (3 images; Figure 7)**

```
Candidate DESI-091.7214-58.9787 --- grz cutout, 26.5" field. Views follow:
[full] whole cutout, 26.5" across (~0.26"/px), Lupton-RGB (z=R, r=G, g=B). Lens galaxies are red/orange;
    lensed sources are typically blue.
[zoom] 2.5x center crop --- the 1-5" region where arcs/counter-images appear.
[residual] lens-light removed (per-band minus a smoothed model): low-surface-brightness arcs and counter-
    images stand out against a flat background.
```



full zoom residual

**Figure 7: Views the agent inspected** for DESI-091.7214−58.9787 (consensus A; this episode: agent grade B, $p_{\text{lens}} = 0.76$). Identical render to the tool output returned to the agent (101 px grz cutout, 0.262″/px, Lupton RGB).

**thinking (summarized) — after inspecting the views**

```
Let me analyze the three views carefully:

**Full view (26.5" field):**
- The field shows a central galaxy that appears reddish/orange, surrounded by several other galaxies
- There's a slightly blue/whitish feature visible near the central red galaxy --- in the upper-left area of
     the central object
[...]
**Zoom view (center crop, 1-5" region):**
[...] There's a blue/cyan feature to the upper-left of the red core that could be a lensed arc or blue
    companion galaxy, appearing somewhat diffuse without an obvious counter-image on the opposite side

**Residual view (lens light removed):**
- The residual image reveals a striking blue ring or halo structure surrounding a bright central source
    [...] --- this is strong evidence for gravitational lensing [...]
- The lens-light removal has exposed this blue ring/arc that was harder to see in the full view
[...]
Now I'm evaluating this against the five lensing criteria. The blue source is clearly present in both the
    zoom and residual views, so that scores around 7-8. [...] Actually, let me reconsider the grading
    criteria more carefully. For an A grade, I need strong evidence from at least two of curvature,
    counter_images, or arc_morphology, plus no decisive contaminants. Looking at what the residual shows:
    the blue emission curves around the central red galaxy (strong curvature evidence), and it's clearly
    extended and ring-like rather than point-like (strong arc morphology evidence). A ring structure would
     imply counter-images exist, giving moderate support there too. [...] This could genuinely be an A
    grade, though the noisy imaging makes me hesitant to commit fully. [...] Finalizing this as a **B**
```

```
     grade with moderate confidence around 0.65 and a lens probability of 0.75-0.80, escalating to human
     review given the ambiguity about whether this is a complete symmetric ring or just a nearby blue
     companion.
```

**agent — final grade (JSON)**


```
{
  "grade": "B",
  "criteria": {"blue_source": 7, "low_surface_brightness": 6, "curvature": 7,
               "counter_images": 6, "arc_morphology": 8},
  "p_lens": 0.76,
  "confidence": 0.63,
  "contaminant": null,
  "escalate_to_human": true,
  "rationale": "The residual view is the key diagnostic: after lens-light subtraction a clear cyan/blue
     extended halo appears encircling the central orange-red DEV galaxy, consistent with a partial-to-
     complete Einstein ring. The zoom view corroborates a blue feature offset ~1--2\" to the upper-left of
     the red core (criterion 1 and 3). Arc morphology is the strongest signal --- the blue emission is
     diffuse and elongated rather than point-like, and its curvature tracks the centre. Counter-images
     cannot be cleanly resolved into distinct opposing arcs at this resolution, preventing an A; the ring-
     like envelope in the residual is suggestive but ambiguous between a near-complete ring and an LSB blue
      companion. No spiral, stellar halo, or tidal-tail contaminant is apparent. Escalated for human review
      given the A/B borderline."
}
```


The reasoning narrates what the structured output alone cannot: a per-view visual analysis, criterion-by-criterion scoring, and an explicit A-versus-B deliberation that ends in escalation. Two further reasoning-instrumented recordings of the same system returned B ($p_{\mathrm{lens}} = 0.72$) and B ($p_{\mathrm{lens}} = 0.62$) — stable at B, bracketing the frozen benchmark run's B/0.75.

### A.2 Neutral: an ambiguous Grade-C "possible"

DESI-030.9742+12.4683 — consensus grade C, CNN $p_{\mathrm{meta}} = 0.898$; the benchmark lean run also graded it C ($p_{\mathrm{lens}} = 0.12$). The task message is analogous to A.1.

**tool call — fetch_cutout**

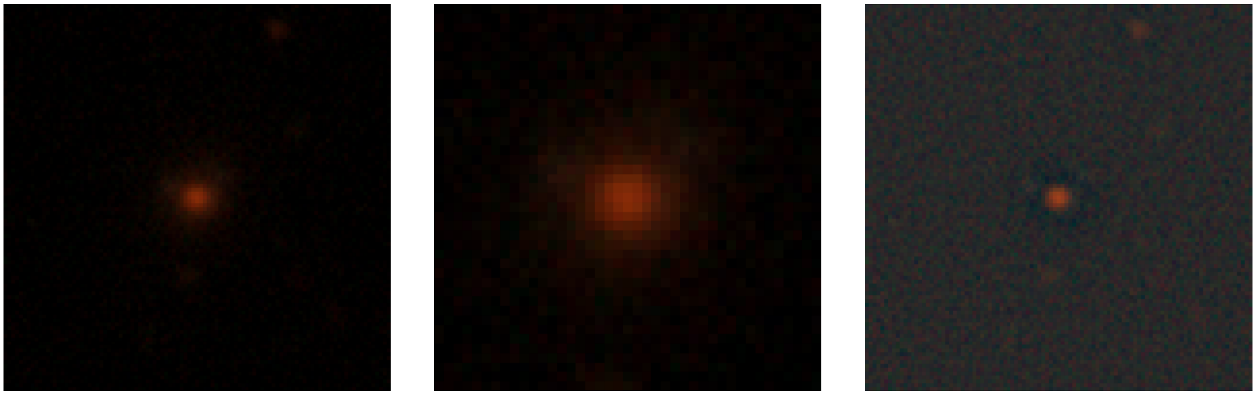

```
{"name": "DESI-030.9742+12.4683", "survey": "storfer", "views": ["full", "zoom", "residual"]}
```


**tool result — fetch_cutout (3 images; Figure 8)**

```
Candidate DESI-030.9742+12.4683 --- grz cutout, 26.5" field. Views follow:
[full] [...] [zoom] [...] [residual] [...]
```

**thinking (summarized) — after inspecting the views**

```
Let me carefully analyze the three views:

**Full view:**
- Central red/orange elliptical galaxy dominating the field
- No obvious blue arcs, rings, or counter-images visible around the galaxy
[...]
**Residual view:**
After subtracting the lens model, I can see the central core remaining---likely indicating incomplete
     subtraction or a ring structure---but the faint features that emerge are reddish/orange rather than
     blue. [...]

**Visual assessment against lensing criteria:**
```



full zoom residual

**Figure 8: Views the agent inspected** for DESI-030.9742+12.4683 (consensus C; this episode: agent grade C, $p_{\mathrm{lens}} = 0.12$, escalated). A smooth red elliptical with a faint annular excess visible only in the residual.

```
The blue source is essentially absent---I don't see any clear blue feature within 1-5" of the central red
     galaxy, and the visible residual features are all reddish/orange. [...] The CNN score is quite high at
      0.898, but visually this looks like an isolated red elliptical galaxy with possibly some ring-like
     residual that could just be a PSF or model subtraction artifact rather than a genuine lensing
     signature.

Looking more closely at the residual, there's a faint ring or partial ring structure around the core, but
     the color is distinctly orange/reddish rather than blue. This could be a very faint Einstein ring or
     just an imperfect model subtraction. Given the lack of clearly blue features, no tangential arc, and
     the residual showing mostly orange/red tones, this is most likely a Grade D or at best Grade C---the
     ring might hint at some lensing, but it's not convincingly blue. [...] The high CNN score creates some
      tension with what I'm seeing visually, which makes this a borderline case worth flagging for human
     review. [...] I'm settling on C with a p_lens around 0.18 and escalating to human review, though the
     isolated red elliptical explanation seems most plausible given the ambiguous residual features.
```

**agent — final grade (JSON)**


```
{
  "grade": "C",
  "criteria": {"blue_source": 2, "low_surface_brightness": 3, "curvature": 1,
               "counter_images": 1, "arc_morphology": 1},
  "p_lens": 0.12,
  "confidence": 0.62,
  "contaminant": null,
  "escalate_to_human": true,
  "rationale": "All three views show a smooth, isolated red elliptical (consistent with the SER Tractor
     type) with no convincingly blue feature at 1--5\" separation in either the full or zoom frames. The
     residual reveals a faint ring-like halo around the subtracted core, but its color is orange-red rather
      than blue, suggesting an imperfect lens-light model fit rather than a lensed source; no tangential
     curvature or arc morphology is identifiable. The single nearby extended object visible to the upper-
     right in the residual is also red/orange, ruling it out as a lensed source. Despite the high CNN prior
      (p_meta=0.898), the absence of any blue arc, counter-image, or tangential elongation reduces p_lens
     substantially; flagged for human review given the discrepancy between the ML score and the visual
     evidence."
}
```


The reasoning shows the C/D boundary being weighed in real time — "most likely a Grade D or at best Grade C" — before settling on the escalated "possible". Two further recordings of the same system landed on the other side of that line (D, $p_{\mathrm{lens}} = 0.07$ and D, $p_{\mathrm{lens}} = 0.12$): the same evidence reads as "possible" or "rejected" on different runs, which is precisely the soft boundary §11 quantifies.

### A.3 Negative: a Grade-D human-rejected contaminant

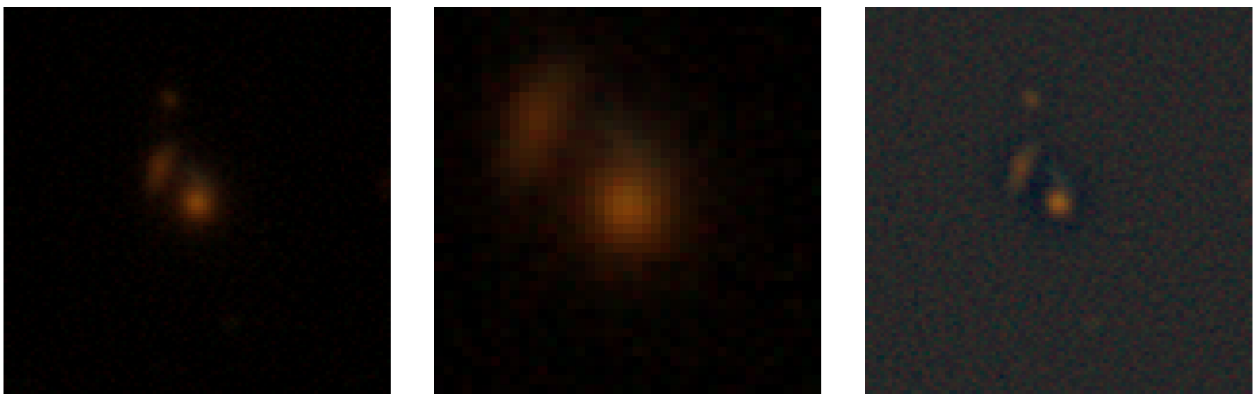
`DESI-124.2632+17.1177` — consensus grade D (human-rejected), yet CNN $p_{\text{meta}} = 0.91$; the benchmark lean run graded it C ($p_{\text{lens}} = 0.32$). The task message is analogous to A.1.

**tool call — fetch_cutout**

```
{"name": "DESI-124.2632+17.1177", "survey": "storfer", "views": ["full", "zoom", "residual"], "ra":
     124.263204, "dec": 17.117705}
```

**tool result — fetch_cutout (3 images; Figure 9)**

```
Candidate DESI-124.2632+17.1177 --- grz cutout, 26.5" field. Views follow:
[full] [...] [zoom] [...] [residual] [...]
```



full | zoom | residual

**Figure 9: Views the agent inspected** for `DESI-124.2632+17.1177` (consensus D; this episode: agent grade D, $p_{\text{lens}} = 0.05$, contaminant `merger`). A close pair of red galaxies that the CNN scored $p_{\text{meta}} = 0.91$.

**thinking (summarized) — after inspecting the views**

```
Let me analyze the three views I received:

**Full view:**
- There's a central red/orange galaxy (the potential lens)
- There appears to be another orange/red blob to the upper-left of the main galaxy
- No obvious blue features are visible - both features appear orange/red in color
[...]
Looking at all three views together, this system appears to be dominated by two red/orange galaxies in
     close proximity with no evidence of blue lensed features, arcs, or a background source - it's more
     consistent with a galaxy group or interaction rather than a strong lensing system. The CNN flagged
     this with a high score of 0.910, but that's a classic false positive---red galaxy pairs can easily
     trigger the model without any actual lensed blue source present. [...] I'm confident enough in the D
     grade and marking this as not needing human escalation.
```

**agent — final grade (JSON)**

```
{
  "grade": "D",
  "criteria": {"blue_source": 1, "low_surface_brightness": 2, "curvature": 0,
```

```
                "counter_images": 0, "arc_morphology": 0},
  "p_lens": 0.05,
  "confidence": 0.82,
  "contaminant": "merger",
  "escalate_to_human": false,
  "rationale": "All three views reveal a red/orange galaxy pair in close angular proximity (~2--3\") with
     no blue source, arc, or tangential feature at any contrast level. The zoom confirms both components
     are similarly red/orange --- consistent with two early-type galaxies at similar redshifts or a galaxy
     group rather than a lens+background-source configuration. The residual view shows only orange over-
     subtraction artifacts and a faint reddish satellite; no blue LSB emission, curvature, or counter-image
      is detected. The high CNN score (p_meta=0.910) is a false positive driven by the compact red galaxy
     pair morphology."
}
```

The shortest reasoning of the three episodes (2,051 summarized characters versus 4,220 for A.1): the contaminant is recognized at a glance — "that's a classic false positive — red galaxy pairs can easily trigger the model" — and the agent declines to escalate. A second recording reproduced D/$p_{\mathrm{lens}} = 0.05$ exactly. The confident rejection with a named contaminant is the pre-filter behavior the report argues is the agent's useful role on ground-based imaging.

# B The cascade in action: two v3 worked examples

This appendix traces the **v3 cascade** (§18) on two candidates to make its two kinds of step visible. *Stage 1* is **deterministic**: `grader_direct` renders the *grz* views and computes aperture photometry *in Python*, then makes a *single* base Messages-API call with the images inline — one turn, no tools, no planning. *Stages 2–3* are **agentic**: the Claude Agent SDK loop, where the model plans its own tool calls (`fetch_cutout`, `get_photometry`, `fetch_hsc_cutout`) across turns. Both were recorded with full message-stream capture (`eval/run_appendix_v3.py`; `claude-sonnet-4-6`, adaptive thinking, the v2 rubric). The three editing rules of Appendix A apply: elision-only (`[...]`), the harness `ToolSearch` tool-discovery call and the `mcp__lens__` namespace prefix elided, and the tool's PNG views reproduced as figure strips.

## B.1 The full cascade: a DESI-buried lens recovered at HSC

`DESI-029.0755−01.1297` — CNN $p_{\mathrm{meta}} = 0.951$, a SuGOHI committee grade-A lens. Both DESI stages ($1.3''$) grade it *D*; only the HSC tier-2 ($0.168''$) flips it to *B*.

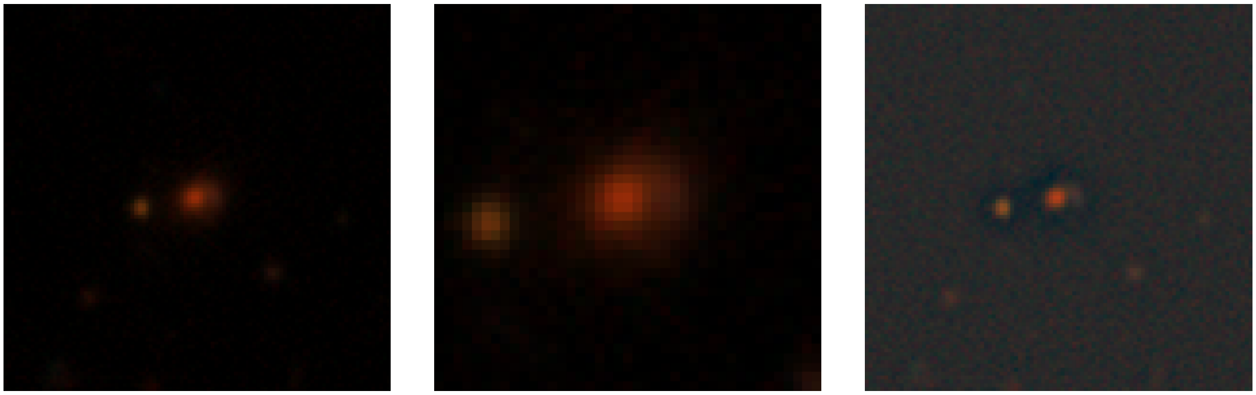
**Stage 1 — deterministic triage (`direct`: render + photometry + one inline call; 1 turn, no tools, \$0.024).** The Python pipeline assembles the evidence; the model is called exactly once.

**deterministic pipeline (Python; no model) — evidence for one inline call (Figure 10)**

```
name='DESI-029.0755-01.1297', survey='storfer', RA=29.075515, Dec=-1.129749.  CNN p_meta=0.951.
render grz views        -> [full, zoom, residual]   (inlined as image blocks)
aperture photometry     -> {"core":    {"g_minus_r": 1.418, "r_minus_z": 1.181},
                             "annulus": {"g_minus_r": 1.268, "r_minus_z": 1.042},
                             "annulus_bluer_than_core": true}
```

**thinking (summarized)**

```
Full/zoom: a central red/orange elliptical with a companion to its left, also reddish. [...]
Photometry: the core is extremely red (g-r=1.418); the annulus is only marginally bluer (g-r=1.268)
-- still far too red to be a lensed blue source. Contaminant: a classic LRG with a red companion --
```



full zoom residual

**Figure 10: DESI $grz$ views** (1.3″ seeing) rendered for `DESI-029.0755−01.1297`: a red core with a red companion that both DESI stages read as an LRG+companion (grade D). The blue arc is blended into the lens light at this resolution.

```
round, compact, no arc, no counter-image, no tangential elongation. Despite the high CNN score, a
textbook false positive.
```

**single inline judgment — final grade (JSON)**

```
{"grade": "D", "p_lens": 0.04, "confidence": 0.88, "contaminant": "lrg_companion",
 "escalate_to_human": false,
 "rationale": "[...] both core (g-r=1.42) and annulus (g-r=1.27) are extremely red; the annulus is
 only marginally 'bluer' by ~0.14 mag, consistent with a less massive red companion rather than a
 lensed blue source. [...] Classic LRG+companion false positive despite the high CNN score."}
```

**Stage 2 — agentic adjudication on DESI (the SDK loop; 4 turns, \$0.134).** Now the model plans its own tools. Here `fetch_cutout` failed to load this candidate's cutout, so the agent fell back to photometry and still concluded D — consistent with stage 1, whose deterministic render (strip above) already supplies the image-based verdict.

**tool call — fetch_cutout**

```
{"name": "DESI-029.0755-01.1297", "survey": "storfer", "views": ["full", "zoom", "residual"]}
```

**tool result — fetch_cutout (ERROR)**

```
ERROR: could not load a cutout for DESI-029.0755-01.1297 (survey=storfer).
```

**tool call — get_photometry**

```
{"name": "DESI-029.0755-01.1297", "survey": "storfer", "ra": 29.075515, "dec": -1.129749}
```

**tool result — get_photometry**

```
{"core": {"g_minus_r": 1.418, "r_minus_z": 1.181}, "annulus": {"g_minus_r": 1.268, "r_minus_z": 1.042}, "
    annulus_bluer_than_core": true}
```

**thinking (summarized)**

```
Cutout failed to load -- I cannot inspect morphology, so photometry is the sole diagnostic. Both core
(g-r=1.42) and annulus (g-r=1.27) sit firmly in red-galaxy color space; a genuine lensed blue arc
would drive the annulus to g-r below ~0.7, not 1.27. The ~0.15 mag offset is the outward color
gradient of a red elliptical or a red companion. [...] Grade D, lrg_companion, escalate for human
review since the image failed.
```

**agent — final grade (JSON)**

```
{"grade": "D", "p_lens": 0.05, "confidence": 0.52, "contaminant": "lrg_companion",
 "escalate_to_human": true,
 "rationale": "Cutout retrieval failed [...] both core AND annulus sit firmly in red-galaxy color
 space -- a genuine lensed blue arc would drive the annulus to g-r below ~0.7, not 1.27. [...]
 LRG+companion strongly favored by the photometry; human review with working image access required."}
```

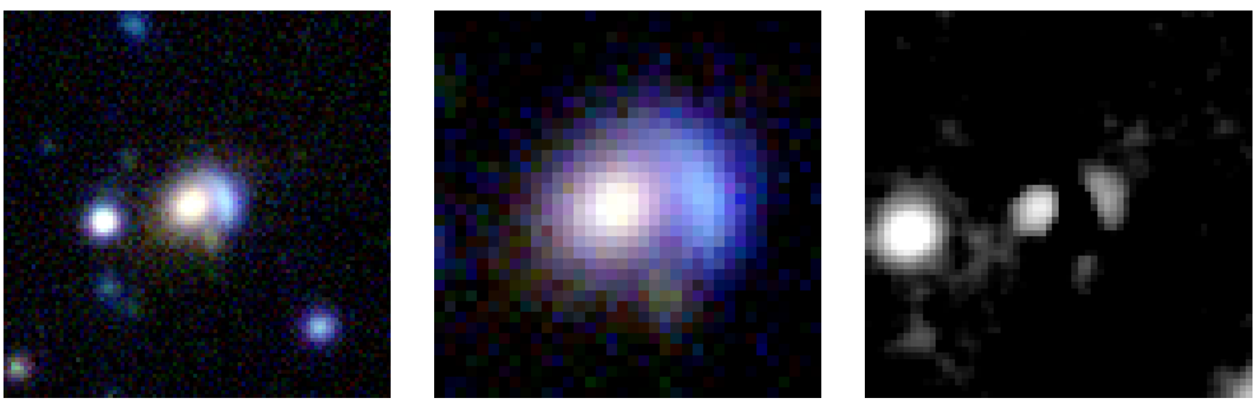
**Stage 3 — agentic tier-2 at HSC 0.168″ (the SDK loop; 3 turns, $0.081).** Coverage resolves to HSC; the agent fetches the grizy cutout and re-grades the same sky position.

**tool call — fetch_hsc_cutout**

```
{"ra": 29.075515, "dec": -1.129749, "views": ["full", "lum", "zoom", "lum_zoom", "lum_sub"]}
```

**tool result — fetch_hsc_cutout (5 images; Figure 11)**

```
HSC-SSP PDR3 ra=29.075515, dec=-1.129749 -- 0.168"/px grizy (bands present: giryz). Views follow:
[full]    16" HSC color (R=i, G=r, B=g), ~8x sharper than DESI grz.
[zoom]    6" center crop -- the arc / counter-image region.
[lum_sub] i-band with the smooth lens light subtracted -- a ring/arc residual stands out.
```



full (color) | zoom (color) | *i*-band, lens-subtracted

**Figure 11: HSC PDR3 grizy views** (0.168″) for the same sky position: the 6″ zoom resolves blue emission wrapping ∼200–270° around the red core, and the lens-subtracted *i*-band shows the lensed-image residual blobs — invisible at DESI's 1.3″. The agent grades this B.

**thinking (summarized) — after inspecting the HSC views**

```
Zoom color (6"): clear blue emission wrapping ~200-270 deg around the red core -- distinctly bluer
than the lens, compelling. lum_sub residual: 2-3 distinct bright blobs at different azimuths after
lens subtraction -- consistent with multiple images or a broken Einstein ring. The blue wrap rules
```

```
out an LRG+companion (same color) and a ring galaxy (color would match). Two of three A-criteria met;
counter-images not fully resolved at ~0.6" seeing, so B. [...]
```

**agent — final grade (JSON)**


```
{"grade": "B",
 "criteria": {"blue_source": 8, "low_surface_brightness": 6, "curvature": 7,
              "counter_images": 6, "arc_morphology": 7},
 "p_lens": 0.62, "confidence": 0.65, "contaminant": null, "escalate_to_human": true,
 "rationale": "The 6\" zoom shows blue emission wrapping ~200-270 deg around the red core, clearly
 bluer than the lens [...] the lum_sub reveals 2-3 distinct blobs consistent with multiple lensed
 images or a broken Einstein ring. Two of three A-criteria met; the blobs are not fully resolved into
 clean arcs at HSC ~0.6\" seeing, keeping the grade at B."}
```

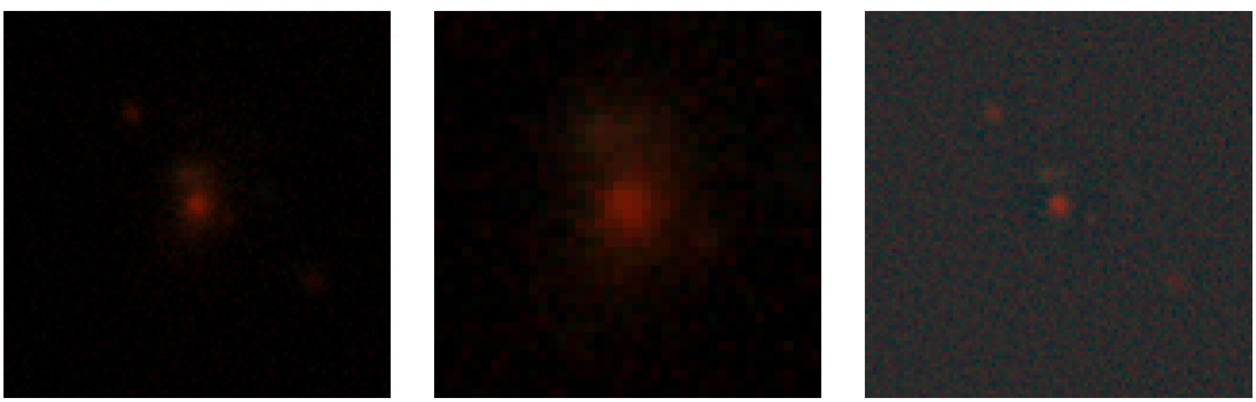

The two DESI stages agree (D, $p_{\text{lens}}\approx 0.04$); only the $0.168''$ tier-2 resolves the blue tangential arc the $1.3''$ pixels had blended into the lens light, flipping the verdict to B. This is the resolution lever of §17 in a single candidate — and the 23/25 recovery of Fig. 5 is this episode, 25 times over.

## B.2 A confident mimic the cheap stage rejects

s_358314_2329 — a ClaudeNet-v3 seed mimic (CNN $p_{\text{meta}} = 0.711$; an agent-confirmed non-lens). The deterministic stage 1 alone settles it; no agentic spend is incurred.

**deterministic pipeline (Python; no model) — evidence for one inline call (Figure 12)**

```
name='s_358314_2329', survey='claudenet', RA=150.518, Dec=4.738.  CNN p_meta=0.711.
render grz views        -> [full, zoom, residual]   (inlined as image blocks)
aperture photometry     -> {"core":    {"g_minus_r": 1.604, "r_minus_z": 1.374},
                            "annulus": {"g_minus_r": 1.323, "r_minus_z": 1.002},
                            "annulus_bluer_than_core": true}
```



full zoom residual

**Figure 12: DESI** $grz$ **views** for the seed mimic s_358314_2329: a red elliptical flanked by two red companions, no blue arc at any contrast — the deterministic stage grades D / `lrg_companion` with confidence 0.88.

**thinking (summarized)**

```
Full: a central red galaxy with two red companions (upper-left, lower-right); all red. Residual: red
point-like residuals, not a blue arc. Photometry: core g-r=1.60, annulus g-r=1.32 -- both far too red
for a lensed source (which would be g-r below 0.5). Consistent with red companions, not a blue lensed
```

```
source. A classic LRG+companion false positive that triggered the CNN via the flanking red
companions. Grade D.
```

**single inline judgment — final grade (JSON)**


```
{"grade": "D", "p_lens": 0.03, "confidence": 0.88, "contaminant": "lrg_companion",
 "escalate_to_human": false,
 "rationale": "All three views show a red elliptical with two nearby red compact sources; no blue
 features. Photometry confirms both core (g-r=1.60) and annulus (g-r=1.32) are extremely red [...]
 the residual shows only red point-like residuals, not a tangential blue arc. Classic LRG+companion
 false positive."}
```


One deterministic turn, \$0.021, no agentic spend. In the rank-routed cascade (§18) this $p_{\mathrm{lens}}$=0.03 ranks the candidate near the bottom of the survivor pool, below the escalation cut — so it is dropped without ever paying for the agentic loop or a tier-2 fetch. The cheap stage cannot tell the hardest mimics from lenses (that is stage 2's job), but it discards the confident ones — and this agent-confirmed mimic is exactly what the hard-negative export of §19 feeds back to the finder.

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